

A Data-Driven Framework for Evaluating Digital Administrative Service Performance in Indonesian Ports: Integrating Port Service Performance Index and Strategic Quadrant Analysis

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Abstract

Digitalization has transformed port administrative services by enabling electronic processing of vessel arrival authorizations. Although this transition potentially accelerates workflows, evaluating its actual impact remains essential since it directly correlates with port performance outcomes. While existing studies heavily concentrate on physical and operational metrics, the role of digital administrative services remains underexplored. Therefore, this study proposed a Port Service Performance Index to evaluate the administrative performance of vessel arrival authorization across Indonesian ports using operational records extracted from the monitoring-inaportnet digital platform. The proposed framework integrates four response time dimensions, including compliance, efficiency, consistency, and robustness. Then, the composite index is subsequently mapped against service volume using a strategic quadrant analysis. The analysis revealed that operational demand is highly concentrated within a limited number of ports, where the congested port category alone accounts for approximately two thirds of national service volume, highlighting the importance of evaluating administrative performance within its operational context. Furthermore, the proposed framework offers a data-driven support for differentiated policy intervention according to the operational performance characteristics. Benchmark ports represent operational references capable of sustaining high performance under relatively heavy workload, while efficient ports category requires capacity planning to preserve service quality under future demand increases. Conversely, congested ports constitute the highest priority for operational intervention due to high demand, whereas developing ports require organizational strengthening and digital infrastructure enhancement. Overall, the proposed framework extends conventional port performance assessment by integrating administrative service quality with operational demand, thereby providing a practical decision-support approach for monitoring and improving digital port administration. To enhance research transparency, the analytical workflow and source code are publicly available through GitHub repositories: <https://github.com/rifkiwijaya12/inaportnetAnalytics>.

Keywords: Port performance, composite index, digital port, strategic quadrant, decision support

Abstrak

Digitalisasi telah mentransformasi layanan administrasi kepelabuhanan melalui penerapan pemrosesan elektronik terhadap perizinan kedatangan kapal. Meskipun transformasi ini berpotensi mempercepat alur pelayanan, evaluasi terhadap dampak aktualnya tetap diperlukan karena secara langsung berkaitan dengan kinerja pelayanan pelabuhan. Namun demikian, sebagian besar penelitian terdahulu masih berfokus pada indikator fisik dan operasional, sementara evaluasi terhadap kinerja layanan administrasi digital masih relatif terbatas. Oleh karena itu, penelitian ini mengusulkan Port Service Performance Index (PSPI) untuk mengevaluasi kinerja pelayanan administrasi perizinan kedatangan kapal pada pelabuhan di Indonesia dengan memanfaatkan data operasional yang diperoleh dari platform Monitoring Inaportnet. Kerangka evaluasi yang diusulkan mengintegrasikan empat dimensi utama waktu respons, yaitu kepatuhan terhadap Service Level Agreement (SLA), efisiensi, konsistensi, dan ketahanan terhadap keterlambatan ekstrem (robustness). Selanjutnya, indeks komposit yang dihasilkan dipetakan terhadap volume pelayanan menggunakan pendekatan Strategic Quadrant Analysis. Hasil penelitian menunjukkan bahwa permintaan pelayanan terkonsentrasi pada sejumlah kecil pelabuhan, di mana kategori Congested Port mencakup sekitar dua pertiga dari total volume pelayanan nasional. Temuan ini menegaskan pentingnya mengevaluasi kinerja pelayanan administrasi dengan mempertimbangkan konteks beban operasional yang dihadapi. Selain itu, kerangka yang diusulkan menyediakan dasar pengambilan keputusan berbasis data (data-driven decision support) untuk merumuskan intervensi kebijakan yang berbeda sesuai dengan karakteristik operasional masing-masing kelompok pelabuhan. Benchmark Ports merepresentasikan pelabuhan yang mampu mempertahankan kinerja pelayanan tinggi meskipun menghadapi beban kerja yang relatif besar, sedangkan Efficient Ports memerlukan perencanaan kapasitas guna menjaga kualitas pelayanan ketika permintaan meningkat di masa mendatang. Sebaliknya, Congested Ports menjadi prioritas utama dalam intervensi operasional akibat tingginya beban pelayanan, sementara Developing Ports memerlukan penguatan kapasitas organisasi serta peningkatan infrastruktur digital. Secara keseluruhan, kerangka yang diusulkan memperluas pendekatan konvensional dalam penilaian kinerja pelabuhan dengan mengintegrasikan kualitas pelayanan administrasi dan volume pelayanan, sehingga menyediakan suatu kerangka pendukung pengambilan keputusan yang praktis untuk memantau dan meningkatkan kinerja administrasi pelabuhan berbasis sistem digital. Untuk meningkatkan transparansi penelitian, alur kerja analitis dan kode sumber tersedia untuk umum melalui repositori GitHub repositories (<https://github.com/rifkiwijaya12/inaportnetAnalytics>)

Kata kunci: kinerja pelabuhan, indeks komposit, Pelabuhan digital, kuadran strategis, pendukung keputusan

1. Introduction

Maritime transport has become the backbone of international logistics, accounting for more than 80% of global trade by volume and distributing approximately 11 billion tonnes annually (Deggim, Yasnikouski, & Istanbulu, 2023). At the national level, Indonesia's nature as an archipelagic state renders it highly dependent on maritime transportation as a vital economic catalyst, which contributed around 7% of the country's GDP in 2025 (Business, 2026). This sector plays several pivotal roles, including fostering economic equality, reducing logistics costs, and facilitating global supply chains that drive industrial growth. Given these profound economic implications, state intervention in regulating the maritime flows of passenger and freight is imperative in order to guarantee territorial sovereignty, maritime security and safety. Achieving these regulatory goals has increasingly demanded the integration of digital transportation systems within port traffic management.

Currently, digital transformation is an inevitable process required to enhance operational efficiency, since minor delays can escalate operational cost and disrupt entire supply chains. To leverage these digital benefits, the Indonesian government through Directorate General of Sea Transport, at Ministry of Transportation, has launched 'Inaportnet' systems. This program aims to generate an integrated digital platform designed to streamline port clearance and vessel approval processes. Remarkably, the system is expected to reduce processing times from a full day to just 10 minutes (Admin, 2018). While this shift offers a significant improvement, evaluating the actual impact of digital intervention becomes crucial stage, as processing speed directly has a relationship with broader metrics of port performance.

In the extensive literature on maritime transportation, port performance to assess operational efficiency has been measured through container throughput, berth utilization, and terminal productivity. Akinsolu and Krmac (2025) analysed container throughput and port efficiency in Europe and China by utilising data envelopment analysis. The study resulted identification of technical efficiency index affected by governance process and resources allocation. While Yao et al. (2024) study proposed berth utilization model integrated with Geographic Information System and big data as reliable proxy for evaluating port service level. Despite these existing studies heavily concentrates on physical and operational metrics, the role of administrative workflows that potentially cause minor delays with disproportionately large logistical impacts remains underexplored. Additionally, the performance of digital administrative service that govern arrival authorization has also received limited attention.

This study develops a measurement framework that represents port service performance across the Indonesian port network. It captures administrative processing time based on compliance, efficiency, consistency, and a robustness metric that measure extreme delays. Furthermore, by establishing strategic quadrant framework to classify ports, the study introduces a performance - service quantity integration framework. Given this clustering process, this methodology enables evaluation of port efficiency and offers policy interventions, thereby allowing policymakers to allocate resources efficiently based on a ports specific quadrant.

2. Methodology

a. Research Design

To evaluate operational performance and derive data-driven policy recommendation for port management, this study develops systematically analytical workflow structured to transform raw high-frequency dataset into empirical indicator capturing port authorisation performance. The raw data is obtained by deploying a web scrapping algorithm to extract operational records from monitoring-inaportnet website. The scope of study is all vessel arrival notification approval (PKK) records throughout 2025 across Indonesian ports utilizing the digital inaportnet platform. Prior subsequent analysis, the raw dataset undergoes exploratory scrutiny to understand data structural characteristics, identify missing values, and apply data-cleaning protocols to guarantee statistical integrity. The following process includes algorithmic derivation of operational variables, primarily feature engineering is conducted to generate the precise response time.

Furthermore, this study consists of two major part including port performance index development and strategic quadrant analysis. Composite index development phase focus on establishing a mathematical formulation to capture multi-dimensional aspects of port operational efficiency, consistency, and reliability. While quadrant analysis focus on integrating previous composite index to traffic scale, thereby classifying ports into effective intervention priority zones. To facilitate transparency and replicability, the comprehensive source code and detailed workflow documentation developed in this study have been made openly accessible via Github repository - <https://github.com/rifkiwijaya12/inaportnetAnalytics>.

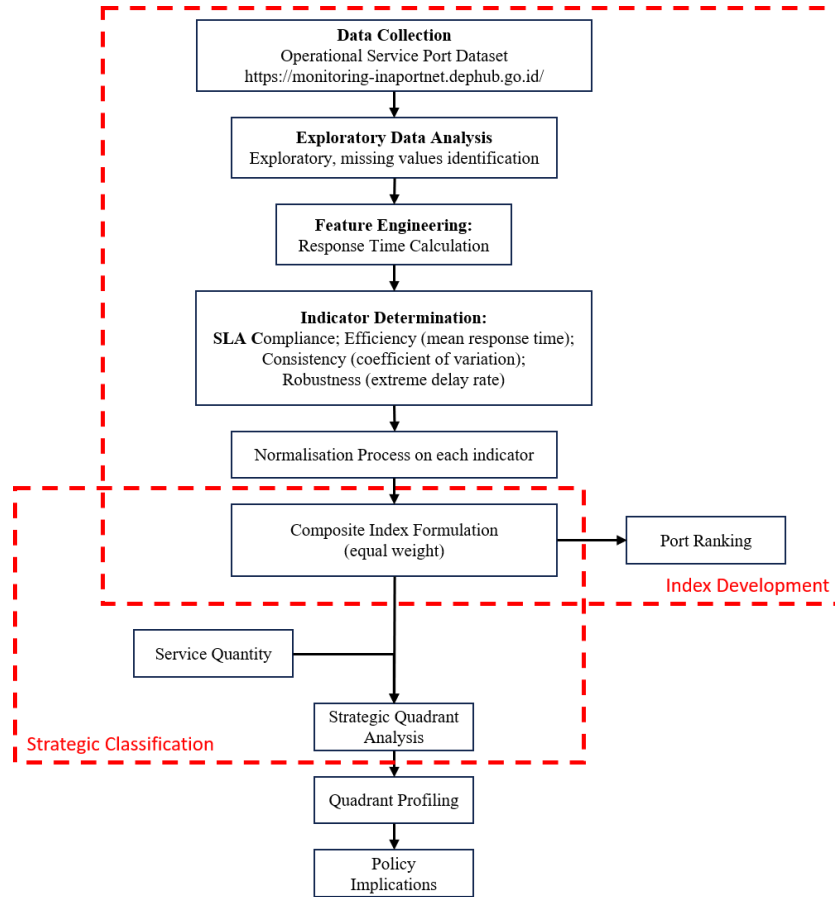


Figure 1 - Research Design Flowchart

b. Port Service Performance Index Development

The primary objective of this stage is to construct a multi-dimensional composite metric, proposed systematically aggregates core operational dimensions, rather than relying on a single response time metric. Since response time alone reflects operational speed, it cannot capture service reliability, compliance with service standards, and consistency across all transactions. Therefore, a multidimensional performance index was developed.

Specifically, it focuses on administrative service performance, which represents the capability of a port authority to process vessel arrival authorizations in a timely, consistent, and reliable manner while complying with established service levels. The standard of approval vessel arrival based on its government regulation (PM 8 Tahun 2022) is maximum of 30 minutes. This process encompasses verifying the completeness and validity of the data and documentation uploaded to the digital application, ensuring strict compliance with regulatory framework before clearance is granted.

Furthermore, a multidimensional performance index consists of four distinct group including compliance, efficiency, consistency, and robustness. Firstly, compliance with service level agreement is developed as a critical metric because adherence to service standards directly improve customer trust to service provider (Shanbhog, Chaudhary, Singh, & Mishra, 2017). By ensuring that port authorities workflows consistently adhere to the standards mentioned in formal regulations, public confidence and trust in

these government institutions can be significantly enhanced. Secondly, efficiency is measured by mean response time refers to the average time taken by an officer respond to submission of vessel arrival. This dimension is often used to evaluate the quality of service in systems such as web services or networked environment ,(Haddad, Mokdad, & Youcef, 2010) which are similar to inaportnet system.

Another metric used in this study is consistency, which is critical for maintaining high quality performance since variation in service leads to customer unsatisfaction (Lee, Lu, Fu, & Teng, 2017). Coefficient of variation is utilised to measure consistency of service data (Bindu, Raghava, Dey, & Rao, 2019). Lastly, robustness index evaluates operational resilience under disruptions by monitoring the frequency of severe delay. Since delay negatively affect in service quality (Parast & Golmohammadi, 2022), thereby, incorporation this dimension into the composite performance index is crucial, as it captures broader view of complexities while standard temporal indicator frequently neglected. The conceptual mapping to establish the core indicators id presented in the Table 1 below:

Table 1 - Conceptual mapping of port service performance indicator

Service Performance Dimension	Operational Indicator	Operational Meaning	Formula
Compliance	SLA Compliance	Ability to satisfy service standards	$CI = \frac{Response\ Time < 30}{Number\ of\ Service}$
Efficiency	Mean Response Time	Processing speed	$EI = \frac{Total\ Response\ Time}{Number\ of\ Service}$
Consistency	Coefficient of Variation	Stability of processing time	$CsI = \frac{Standard\ deviation}{Mean\ Response\ Time}$
Robustness	Extreme Delay Index	Resistance to severe delays	$RI = \frac{Response\ Time > 120}{Number\ of\ Service}$

Because all indicators are generated by different scales and directions of interpretation, data normalisation proses is employed to transform all variables into a uniform scale and can be aggregated into a final composite index as shown in Equation 1. In addition, the normalized indicators represent complementary dimensions of service performance quality, therefore equal weighting was adopted to preserve methodological transparency and interpretability. The resulting port service performance index range between 0 and 1, where higher scores indicate better service performance. Subsequently, all ports are ranked according to their values to facilitate comparative performance assessment.

$$PI = \frac{1}{4}(CI + EI + CsI + RI) \quad (1)$$

Where PI : port performance index;
CI : compliance rate index
EI : efficiency index
CsI : consistency index
RI : robustness on extreme delay index

c. Strategic Quadrant Analysis

Although port performance index enables comparative performance evaluation across all ports, performance ranking provides limited guidance for strategic resource allocation and improvement strategies. Therefore, this study integrates service performance index with service demand (total volume) through strategic quadrant framework to classify effective policy intervention. Given this consideration, performance index is described as a measure of the quality of administrative service, while traffic volume represents as a metric to quantify the operational workload. Median threshold on each dimension is utilised based on its capability to provide accurate and balance classification. Overall, classification of 4 quadrant analysis is presented in Figure 2 with definition as shown in Table 2.

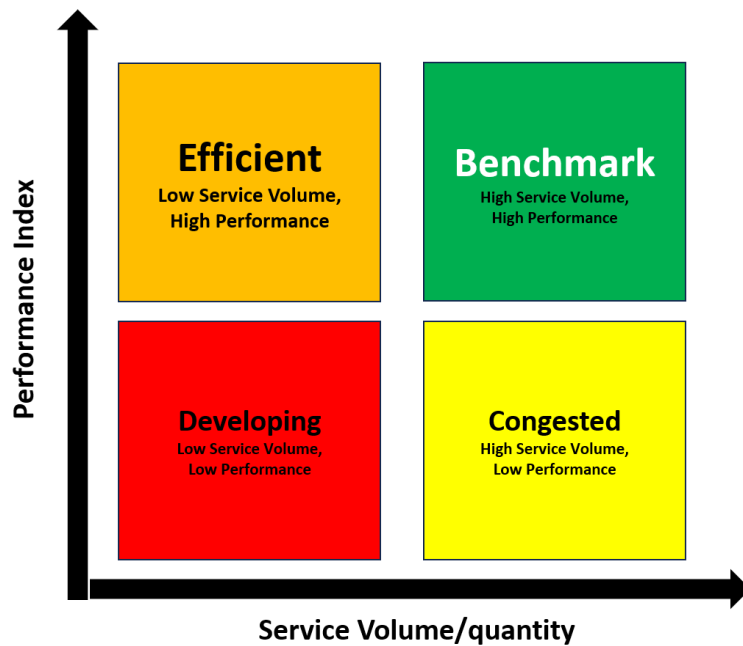


Figure 2 - Quadrant matrix of Port Performance - Service Volume

Table 2 - Quadrant interpretation

Quadrant	Description	General Interpretation	Strategic Focus
Benchmark	High Performance – High Volume	Strong service performance under heavy operational demand	Maintain performance and disseminate best practices
Efficient	High Performance – Low Volume	High service quality with remaining operational capacity	Encourage service utilisation and future growth
Developing	Low Performance – Low Volume	Operational performance requires improvement	Capacity building and process optimisation

		despite manageable workload	
Congested	Low Performance – High Volume	High operational pressure accompanied by poor service performance	Priority operational intervention

The quadrant classification provides a strategic framework for identifying ports with critical prioritisation intervention, different operational context and performance level. Moreover, the resulting classification serves as the basis for subsequent comparative analysis and policy recommendations.

3. Result and Discussion

a. Descriptive analysis

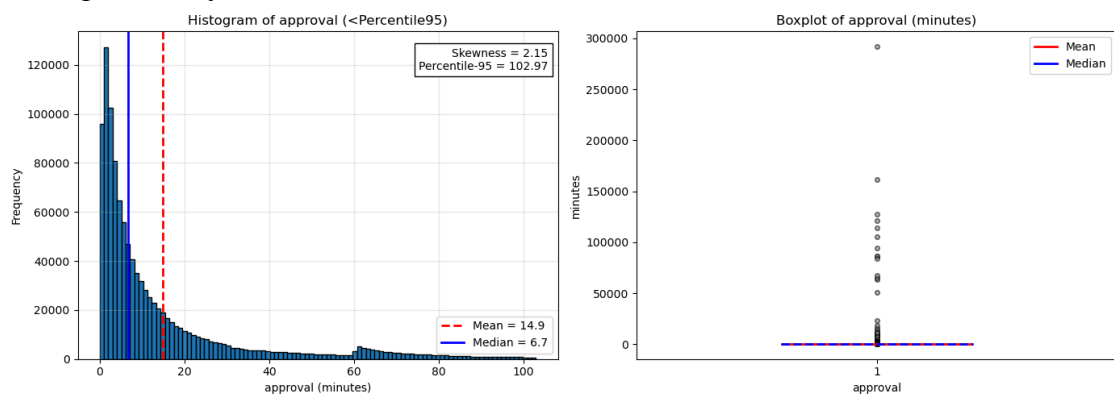


Figure 3 - Histogram and boxplot of approval in minutes across 259 ports

Table 3 - Descriptive statistics of approval in minutes across all ports

Number of Data	Min	Q1	Median	Q3	Max	Mean	Std Dev
1,155,919	0.00	2.7	7.317	20.65	292,161.55	31.491	442.147

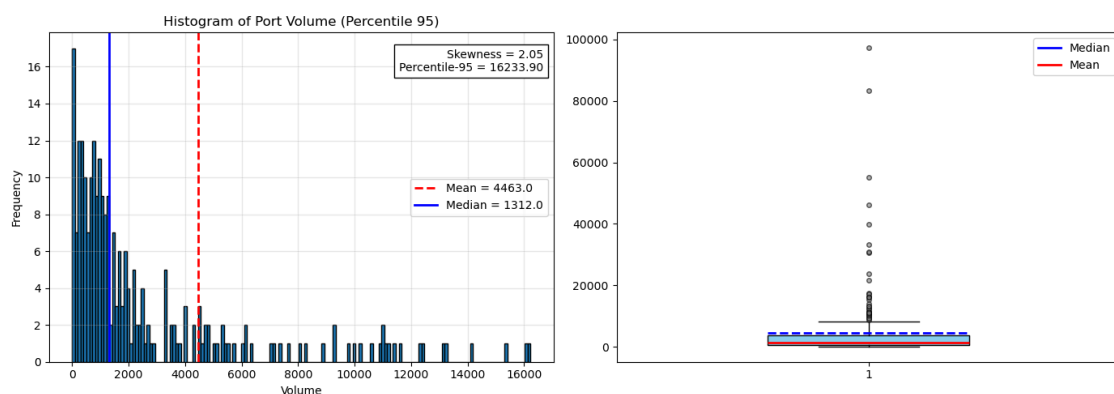


Figure 4 - Distribution analysis of service volume on each port

Table 4 - Descriptive statistics of service volume on each port

Number of Data	Min	Q1	Median	Q3	Max	Mean	Std Dev
259	2.00	609.5	1,312.0	3870,0	97,253	4463.0	10,306.88

As shown in Figure 3, the histogram analysis of approval times less than 8 hours across 259 ports in Indonesia during 2025 revealed a highly right skewed distribution. The data reflected substantial operational variability among dataset of 1,098,115 observations (approximately 95% of all approval transactions). Across all ports, while the median of approval time presented efficient at 7.317 minutes, the mean was significantly elevated to 31.491 minutes with an extraordinarily high standard deviation of 442.147 minutes. This substantial gap between mean and median highlighted the profound impact of extreme outliers, which culminated in the highest approval time of 292,161 minutes (approximately 202 days).

Furthermore, a highly right-skewed distribution was represented in the service volumes managed by 259 ports. This means that the majority of ports handled a relatively small volume of service with the smallest service volume at Kuala Langsa port of 2. On the contrary, a few giant hub ports experienced an extraordinarily large volume, reaching a maximum of 97,253 transaction at Batam port during 2025. The median of service volume stood at a modest 1,312 approvals, while the mean was in a higher position at 4,463 units. Therefore, service quantity across all ports indicated a pronounced heterogeneity in national port services, highlighting that capacity and operational demands remain heavily centralized in a few strategic locations.

To gain deeper understanding of these operational variations, the descriptive analysis is extended to each 4 key indicators, including compliance, efficiency, consistency, and robustness. Firstly, SLA compliance displayed a pronounced negative skewness at -1.85, indicating that while most ports maintained high compliance rates with a median of 0.86, a small portion of underperforming ports significantly pulled the mean downward to 0.80. Conversely, the remaining three dimensions showed distinct patterns. Efficiency, consistency, and robustness dimension presented highly right skewed. Boxplot analysis as shown in Figure 5 presents that all dimensions have extreme outliers. This condition introduces a critical methodological challenge, since standard normalization would heavily bias the scaling process. Therefore, to handle the influence of outliers, winsorized normalization was applied (Abuzaid & Alkrunz, 2024) by capping extreme values at specified percentiles. This approach effectively neutralizes the distorting influence of anomalous administrative lags and outlier hub ports.

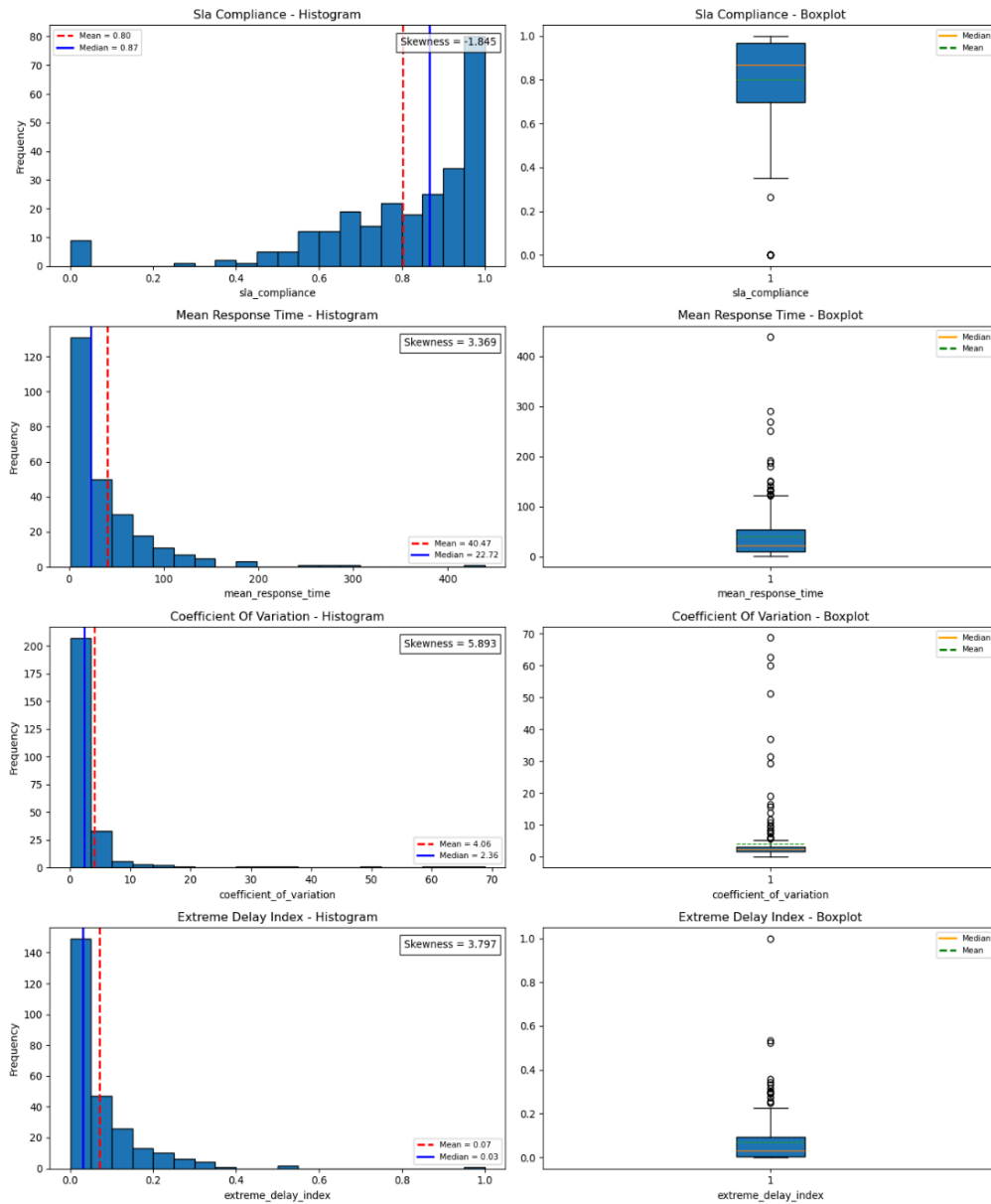


Figure 5 - Distribution analysis of each performance indicator

Table 5 - Descriptive statistics of all performance indicators

Dimension	Indicator	Mean	Median	Std dev	Skewness	Normalization
Compliance	SLA compliance	0.80	0.86	0.22	1.85	Winsorized Min-Max
Efficiency	Mean response time	40.47	22.72	51.52	3.39	
Consistency	Coefficient of variation	4.06	2.36	8.18	5.93	
Robustness	Extreme delay index	0.07	0.03	0.11	3.92	

b. Port Service Performance Ranking

Figure 6 presents that the majority of Indonesian ports achieved performance index above 0.76, indicating generally satisfactory administrative approval performance, although significant operational divergence is still observed across ports. The final

aggregation of performance metrics yields a moderate left-skewed of composite index distribution. The overall performance exhibits a clear concentration toward the higher scale with median index of 0.804 and third quartile reached at 0.906, these values indicated that a substantial proportion of national ports operate with relatively high efficiency and compliance. Despite a low minimum index score of 0.052, the low variability indicated by a standard deviation of 0.188 confirms that the implementation of the Winsorized Min–Max normalization successfully stabilized the final index, preventing extreme operational anomalies from over-compressing the scores.

Furthermore, in depth analysis of highest and lowest rankings highlights that there is a distinct trade-off between a port operational workload and its composite performance score, as detailed in Tables 6 and 7. The top 5 best performing approval namely Nabire, Tg. Sarang Elang, Bawean, Indramayu, and Manggar, consistently achieve perfect composite scores or close to 1. The perfect composite index is a direct reflection of the optimal performance scores demonstrated across all individual sub-indicators. Such perfect ratings are directly enabled by the relatively low to moderate service demands, ranging from only 28 to 1,877 submissions during the year 2025, which averages of only around 5 submission per day. Conversely, the bottom five lowest performance ports including Satui, Pangkal Balam, Padang Bai, Molawe, and Kuala Semboja, exhibit severely depressed composite scores with a minimum of 0.0516. Notably, all five underperforming ports handle substantially larger operational workloads, typically exceeding 5,00 to around 17,000 transactions. This substantial divergence clearly demonstrates that elevated service volumes impose a bottleneck on administrative approval, leading to a critical breakdown in compliance and robustness metrics. Therefore, this combination of high volume and low performance index highlights the necessity of capacity scaling and process optimization.

While this port performance index ranking offers an initial overview of each port's position, developing targeted intervention strategies requires consideration of service volume. Further evaluation of volume and performance metrics will be conducted in the subsequent section

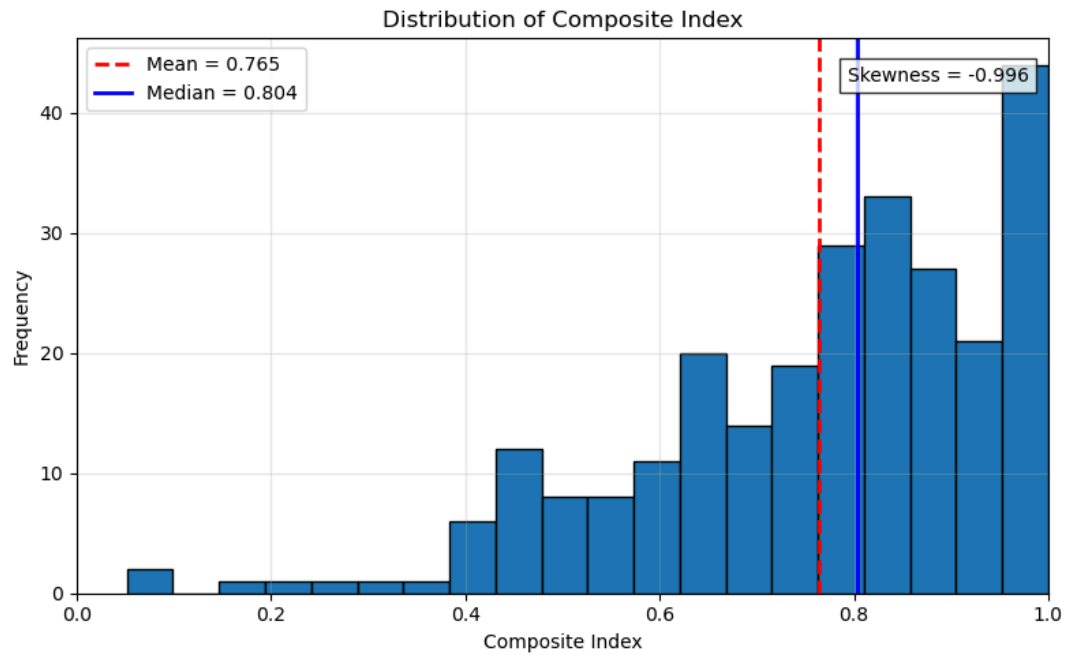


Figure 6 -Distribution of Composite index across all ports

Table 6 - Top 5 best performance index

Port	Volume	Compliance index	Efficiency index	Consistency index	Robustness index	Composite index
NABIRE	703	1	1	1	1	1
TG. SARANG ELANG	28	1	1	1	1	1
BAWEAN	922	1	1	1	1	1
INDRAMAYU	1877	1	1	1	1	1
MANGGAR	1201	1	0.9995	1	1	0.9998

Table 7 - Bottom 5 lowest performance index

Port	Volume	Compliance index	Efficiency index	Consistency index	Robustness index	Composite index
SATUI	8085	0	0.0465	0.9813	0	0.2569
PANGKAL BALAM, BANGKA	5323	0.0161	0	0.8601	0	0.2190
PADANG BAI	10960	0	0.2879	0	0.3329	0.1552
MOLAWA	17487	0	0.2568	0.0115	0	0.0671
KUALA SEMBOJA	10577	0.0390	0	0.1671	0	0.0516

c. Strategic Quadrant Classification

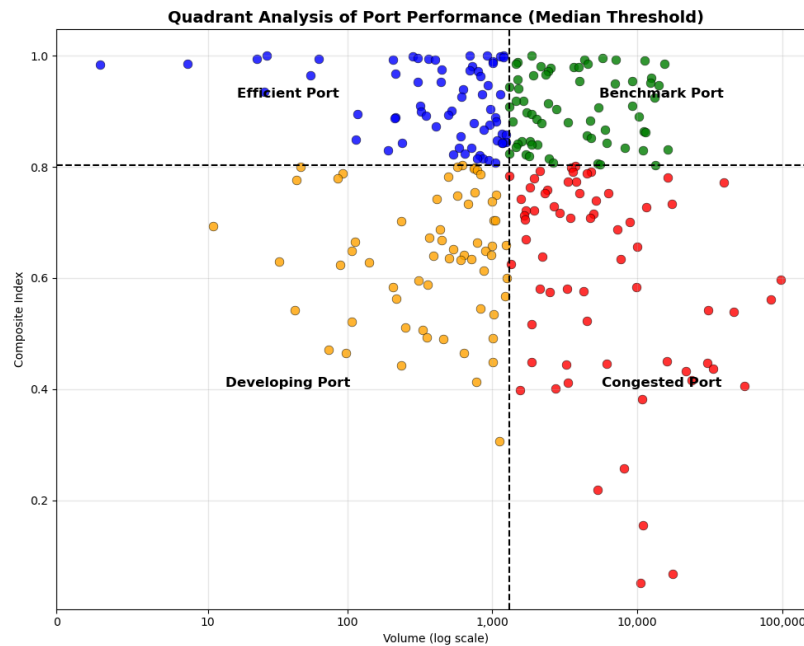


Figure 7 - Scatter Plot of quadrant analysis based on median threshold

Figure 7 presents quadrant classification derived from the intersection of median values of volume distribution and performance index. Due to the highly right skewed distribution of service quantity across all Indonesian ports, a logarithmic scale was applied to improve visual clarity. This strategic quadrant analysis aims to classify ports into distinct managerial strategies. Rather than evaluating performance in isolation, this approach incorporates operational context represented by service volume. As detailed in Table 8, median-based classification resulted in a relatively balanced distribution of ports across the four quadrants, which each quadrant containing approximately 25% of total ports. This balanced distribution is a direct mathematical consequence of utilising median as the classification threshold.

Furthermore, the quadrant analysis reveals that major gateway such as Tanjung Priok, Jakarta and Tanjung Perak, Surabaya serve as benchmark ports due to their ability to maintain high performance indices despite substantial traffic volumes. Similarly, on the opposite end of the volume scale, Kuala Langsa and Teluk Leidong port consistently secure their positions as efficient ports. However, low service volume does not inherently guarantee the efficiency, as represented by Sirombu ports, which falls under developing ports despite handling only 12 transactions in 2025. In addition, a systemic vulnerability is observed at the upper extreme, where the ten largest ports by volume are heavily concentrated in the congested quadrant, indicating that escalating transaction volumes severely compromise their ability to sustain optimal performance indices.

While the number of ports is evenly distributed across all quadrant, ranging from 63 to 66 ports per quadrant, the actual workload is heavily concentrated in congested port. This quadrant handles 65.32% of the national volume, recording the highest average of service volume with a mean volume of 11,439.24. Consequently, this severe workload correlates with a degraded efficiency, as evidenced by its low mean performance index of 0.60. In contrast, Benchmark ports are able to manage a substantial mean volume of 5,071.19 (28.08% total share) from performance

degradation, maintaining a high mean performance index of 0.90. Meanwhile, efficient and developing ports experience minimal operational pressure, collectively managing less than 7% of the total share volume. Efficient ports maximize their low volume with the highest mean performance index of 0.92, while developing ports demonstrate inability to capitalize on relatively low workload with mean volume of 553.51, resulting in a lower mean performance index of 0.64.

Table 8 - Port Classification

Classification	Number of Ports	Mean Performance Index	Mean Volume	Total Volume	% Share Volume
Benchmark Port	64	0.90	5071.19	324,556	28.08%
Congested Port	66	0.60	11439.24	754,990	65.32%
Developing Port	63	0.64	553.51	34,871	3.02%
Efficient Port	66	0.92	628.82	41,502	3.59%

d. Discussion

Although the average of performance index showed acceptable administrative efficient performance, the high variability among individual ports highlight that digital service quality has not been uniformly realized across the national port infrastructure. The findings highlight port performances experience highly divergent patterns relative to service. While several port with high transaction volumes can preserve optimal service standard, others suffer from severe efficiency degradation since transaction volumes escalate. In contrast, a significant number of low service quantity displayed underperformance, highlighting that low operational demand is insufficient to ensure satisfactory service quality, particularly within compliance dimension which requires closer attention.

Furthermore, benchmark port category represents the most desirable operational condition within the proposed network. This finding suggests that maintaining service performance under heavy operational workload can be achieved. Governance practice, resource allocation effectively, or operational procedures may contribute to ability of some ports to sustain high approval performance. Future studies may integrate organizational characteristics to explain why certain ports are capable of maintaining consistently high performance. While efficient port category demonstrates operational efficiency, the scalability of their current performance under escalating traffic demand remains highly questionable. Hence, this category requires strategic capacity planning to ensure that operational efficiency can be maintained as transaction volumes scale up. In contrast to high performing quadrants, congested and developing ports category represent critical areas that require policy intervention to improve service approval performance. In particular, the congested ports indicate that increasing the quantity of service may exceed existing capacity, resulting in average longer response and ultimately reduced overall performance. Several Indonesia's busiest ports (top 10 service volume) fall within this category. This finding align with previous study (Bolat, KayiŞoğlu, Güneş, Kizilay, & Özsöğüt, 2020) indicating that increased traffic volume is a primary cause for port congestion which ultimately impacting operational port performance. For instance, Batam port which recorded the highest overall volume within the entire network, handling six times the traffic of Tanjung Priok port which the

leading service volume in the benchmark category. Such contrast comparison suggests that multifaceted approach should be conducted through workflow redesign, operational optimization, and proportional expansion of human resources. Unlike congested port, the developing port category is likely to occur in remote ports such as Agats, Pulau Telo, and Sinabang, illustrate how geographical or infrastructure isolation may constrain digital administrative service. Conversely, newly established strategic port such as Patimban may still be undergoing institutional and operational adjustments before reaching mature performance levels.

Therefore, the proposed framework provides structured decision support tool for authorities in supporting continuous performance monitoring and evidence-based policy formulation within the Inaportnet administrative system. As summarized in Table 9, each quadrant has a specific policy intervention based on operational characteristics.

Table 9 - Proposed strategic policy intervention

Quadrant	Strategic Objective	Priority Intervention
Benchmark Ports	Preserve operational excellence	Continuous performance monitoring, dissemination of best practices, knowledge sharing
Efficient Ports	Prepare for future service growth	Capacity planning, workforce preparedness, preventive process optimization
Congested Ports	Restore service efficiency	Process reengineering, workload balancing, additional staffing, managerial intervention
Developing Ports	Strengthen organizational capability	Capacity building, digital infrastructure improvement, technical mentoring, competency enhancement

This study has several limitations that should be considered when interpreting the findings. Firstly, the analysis was conducted based on digital approval system during 2025, where several ports represent an early stage of implementation. Consequently, the observed performance may partially reflect differences in adoption and operational maturity. Secondly, the service performance evaluations exclusively utilised by PKK approval within inaportnet systems. Ports that continue to rely partly on manual administrative procedures may not be fully represented, also another service such as SPM/SPB (entrance/exit approval) or other service therefore not captured in the analysis. Thirdly, although the proposed framework successfully classifies in service performance across all ports, it does not investigate the organizational, technological, or managerial factors responsible for underperformance port. Therefore, future research should advance the proposed framework through multi-year operational dataset, extend validation across diverse digital service, and incorporate explanatory variables to give better understand the determinants of administrative service quality.

4. Conclusion

This study proposed a Port Service Performance Index (PSPI) as a multidimensional framework for evaluating the administrative performance of vessel arrival authorization

across Indonesian ports using operational records extracted from the Monitoring Inaportnet platform. Unlike conventional port performance assessments that primarily emphasize physical infrastructure and operational throughput, the proposed framework integrates four complementary dimensions of administrative service quality, namely compliance with service standards, processing efficiency, consistency, and robustness against extreme delays. Furthermore, integrating the composite index with service volume through Strategic Quadrant Analysis enables performance evaluation within the context of operational workload, thereby providing more meaningful managerial insights than performance ranking alone.

The empirical findings reveal that, although the overall administrative performance of Indonesian ports is generally satisfactory, substantial disparities remain across individual ports. Service demand is highly concentrated within a limited number of ports, with the Congested Port category accounting for approximately two-thirds of the national transaction volume while simultaneously exhibiting the lowest average performance index. In contrast, Benchmark Ports demonstrate that high administrative performance can be sustained even under heavy operational demand, suggesting that effective organizational practices and operational capacity can mitigate the adverse effects of increasing workloads. Meanwhile, Efficient Ports possess considerable capacity for future growth, whereas Developing Ports require improvements in organizational capability and digital infrastructure despite handling relatively low service volumes.

From a practical perspective, the proposed framework offers a data-driven decision support tool for continuous monitoring and evidence-based policy formulation within Indonesia's digital port administration system. By classifying ports according to both service quality and operational demand, policymakers can prioritize interventions that are proportional to each port's operational characteristics, thereby improving the effectiveness of resource allocation and performance management. Beyond the Indonesian context, the proposed methodology can also serve as a reference for evaluating digital administrative services in other transportation and public service sectors where response time, service reliability, and workload must be considered simultaneously.

Despite these contributions, the present study is limited to vessel arrival authorization (PKK) records during a single year of observation and does not explicitly examine the organizational or technological determinants influencing service performance. Future research should incorporate multi-year datasets, expand the evaluation to additional digital maritime services, and integrate explanatory variables related to organizational capacity, human resources, and digital infrastructure. Such extensions would strengthen the understanding of administrative service performance and further enhance the proposed framework as a comprehensive tool for digital port governance.

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